Wireless closed-loop optogenetics across the entire dorsoventral spinal cord in mice

Paper: Kathe et al. (2022 (online 2021)). Mouse spinal-circuit device and intervention study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Practical closed-loop measurement and control of motor circuits. Demonstrates behavioural and muscle outcomes outside the basal ganglia.

How does it answer the general questions?

Measured and controlled. A stretchable epidural micro-LED implant and wireless headstage combine optical access to targeted spinal pathways with physiological recordings. EMG-based triggering enables stimulation at selected movement phases; experiments assess effects on locomotion and muscle activity (PDF pp. 1–10).

Usable insight. Phase-specific activation or silencing can produce distinct motor effects. The control objective is defined by locomotor function and circuit interrogation, not beta power. Extended Data Fig. 3 separates approximately 11.3 ms signal processing from 0.9 ms stimulus-delivery delay in a bench timing test (PDF p. 20).

Model relevance. EMG and movement measurements are plausible observed/performance outputs for a neuromuscular model, with explicit sensing and processing dynamics. They are not direct measurements of basal-ganglia population rates. The paper provides no STN–GPe parameterization; its hardware timing and actuator gains cannot be transferred unchanged to DBS or another optogenetic preparation.

What papers does it cite that could provide more information?

- Montgomery et al. (2015), *Wirelessly powered, fully internal optogenetics for brain, spinal and peripheral circuits in mice*: wireless actuation (ref. 6, PDF p. 11).
- Capogrosso et al. (2018), *Configuration of electrical spinal cord stimulation through real-time processing of gait kinematics*: feedback implementation (ref. 40, PDF p. 14).